Understanding Diabetes

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What is diabetes?

Diabetes mellitus is a chronic condition in which the body cannot properly regulate the level of glucose in the blood. Glucose is the main source of energy for the body's cells. The hormone insulin, made by the pancreas, allows glucose to move from the blood into cells. In diabetes, the body either does not make enough insulin or cannot use insulin effectively.

Types of diabetes

- Type 1 diabetes is an autoimmune condition in which the immune system destroys the insulin-producing beta cells of the pancreas. People with type 1 diabetes need insulin every day.
- Type 2 diabetes is the most common form. The body becomes resistant to insulin, and over time the pancreas cannot make enough to keep blood glucose normal.
- Gestational diabetes develops during pregnancy and usually goes away after delivery, but it raises the mother's risk of developing type 2 diabetes later.
- Prediabetes means blood glucose is higher than normal but not yet high enough to be called diabetes.

Common symptoms

- Increased thirst and frequent urination.
- Extreme hunger and unexplained weight loss.
- Fatigue and blurred vision.
- Slow-healing sores and frequent infections.
- Tingling, pain or numbness in the hands or feet.

How diabetes is diagnosed

A fasting plasma glucose of 126 mg/dL (7.0 mmol/L) or higher on two separate tests indicates diabetes. An HbA1c of 6.5% or higher also indicates diabetes. HbA1c reflects the average blood glucose over the past two to three months. A two-hour glucose of 200 mg/dL or higher during an oral glucose tolerance test confirms the diagnosis. Prediabetes is diagnosed when

fasting glucose is 100 to 125 mg/dL or HbA1c is between 5.7% and 6.4%.

Complications

Long-term high blood glucose damages blood vessels and nerves. Complications include heart disease and stroke, kidney disease (nephropathy), nerve damage (neuropathy), eye damage (retinopathy) that can lead to blindness, and foot problems that may lead to amputation.

Managing diabetes

- Follow a balanced diet that limits sugary drinks and refined carbohydrates.
- Aim for at least 150 minutes of moderate physical activity each week.
- Monitor blood glucose as advised by your healthcare team.
- Metformin is usually the first medicine prescribed for type 2 diabetes. Insulin is required for type 1 diabetes and is sometimes needed in type 2.
- Attend regular check-ups for eye, kidney and foot examinations.

Low blood sugar (hypoglycemia)

Hypoglycemia occurs when blood glucose falls below 70 mg/dL (3.9 mmol/L). Symptoms include shaking, sweating, hunger, dizziness, confusion and irritability. Treat it with 15 grams of fast-acting carbohydrate such as glucose tablets or half a cup of fruit juice. Recheck blood glucose after 15 minutes and repeat if it is still low. Seek emergency help if the person becomes unconscious or cannot swallow.